

Tutorial 1

Maps, Vector spaces and Duality.

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1. Decompose the given map f as a composition of elementary operations, such as squaring. Define each elementary map using the $A \rightarrow B$; $a \mapsto b$ notation.

$$f(x) = \sqrt[3]{\cos(x^2)}, \quad x \in \mathbb{R}$$

For each elementary map, specify whether it is injective, surjective, bijective or none with the domain A and target (codomain) B you defined for the map.

Solution: One possible decomposition is:

$$\begin{array}{ccccccc} \mathbb{R} & \xrightarrow{(\cdot)^2} & \mathbb{R} & \xrightarrow{\cos} & \mathbb{R} & \xrightarrow{\sqrt[3]{\cdot}} & \mathbb{R} \\ x & \mapsto & x^2 & \mapsto & \cos(x^2) & \mapsto & \sqrt[3]{\cos(x^2)} \end{array}$$

Another possibility is:

$$\begin{array}{ccccccc} \mathbb{R} & \xrightarrow{(\cdot)^2} & \mathbb{R} & \xrightarrow{\cos} & [-1, 1] & \xrightarrow{\sqrt[3]{\cdot}} & [-1, 1] \\ x & \mapsto & x^2 & \mapsto & \cos(x^2) & \mapsto & \sqrt[3]{\cos(x^2)} \end{array}$$

One more possibility is:

$$\begin{array}{ccccccc} \mathbb{R} & \xrightarrow{(\cdot)^2} & \mathbb{R}_{\geq 0} & \xrightarrow{\cos} & [-1, 1] & \xrightarrow{\sqrt[3]{\cdot}} & [-1, 1] \\ x & \mapsto & x^2 & \mapsto & \cos(x^2) & \mapsto & \sqrt[3]{\cos(x^2)} \end{array}$$

2. Tick the true statements:

- ☐ Every vector is a matrix with only one column.
- ☐ Every linear map between vector spaces can be represented by a unique quadratic matrix.
- ☒ Every vector space has a unique corresponding dual vector space.
- ☐ Every vector in a vector space has a unique dual covector in the dual space.
- ☒ One can always reconstruct a vector from its components and the corresponding basis.
- ☒ A change of basis in a vector space does not change a vector in that space.
- ☐ Linearity of a map $\varphi: V \rightarrow W$ depends on the choice of basis in V and W .

- ✓ Given a basis of a vector space, one can compute components of covectors in the dual space with respect to the dual basis.
- The length of a vector $(3, 4, 5) \in \mathbb{R}^3$ with the standard vector space structure is $\|v\| = \sqrt{50}$.

3. Notation reminder:

Space	V	V^*
Elements	$v \in V$	$\alpha \in V^*$
Basis elements	e_i	ε^j
Components	$v^i \in \mathbb{R}$	$\alpha_j \in \mathbb{R}$
Decomposition	$v = v^i e_i$	$\alpha = \alpha_j \varepsilon^j$
Matrix notation	$\mathbf{v} = \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix}$	$\boldsymbol{\alpha} = (\alpha_1 \quad \cdots \quad \alpha_n)$

⚠ Pay attention: some people arrange covector components not in a row but in a column. This way, they get expressions in components like $P = \boldsymbol{\alpha}^\top \cdot \mathbf{v}$, where $\boldsymbol{\alpha}^\top$ is now a row. If the components are arranged as a row (as in the table), we have $P = \boldsymbol{\alpha} \cdot \mathbf{v}$. The transformation of components of a covector is written with the row notation (as in the table) as:

$$\tilde{\boldsymbol{\alpha}} = \boldsymbol{\alpha} \cdot A \quad \text{or} \quad \tilde{\boldsymbol{\alpha}}^\top = A^\top \cdot \boldsymbol{\alpha}^\top$$

(the second equation is the transpose of the first one) instead of

$$\tilde{\boldsymbol{\alpha}} = A^\top \cdot \boldsymbol{\alpha}$$

with the column notation. Look at the lecture notes for a detailed explanation/derivation and compare with the slides.

4. Consider the composition $\varphi \circ \theta$ of two linear maps $\theta: U \rightarrow V$ and $\varphi: V \rightarrow W$ with U, V and W having the dimensions k, n and m , respectively. Let $(g_1, \dots, g_l), (e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$ be bases in U, V and W . Show that the coordinate representation of the composition $\varphi \circ \theta$ is given by the matrix product $\Phi \cdot \Theta$ of the coordinate representations Φ of φ and Θ of θ . First, obtain an expression in components and then verify that the matrix multiplication rule ‘row times column’ replicates this expression.

Solution: Suppose that $u = u^i g_i$. Then

$(\varphi \circ \theta)(u) = \varphi(\theta(u))$	definition of composition
$= \varphi(\theta^i_j u^j \cdot e_i)$	definition of components of θ
$= \theta^i_j u^j \cdot \varphi(e_i)$	linearity of φ
$= \theta^i_j u^j \varphi^k_i \cdot f_k$	definition of components of φ
$= (\varphi \circ \theta)^k_j u^j \cdot f_k$	definition of components of $\varphi \circ \theta$.

Therefore, the components of the composition $\varphi \circ \theta$ are given by

$$(\varphi \circ \theta)^k_j = \varphi^k_i \theta^i_j \quad \forall k = 1, \dots, m, \quad \forall j = 1, \dots, l.$$

This can be written using matrix multiplication as

$$\begin{pmatrix} (\varphi \circ \theta)^1_1 & \cdots & (\varphi \circ \theta)^1_l \\ \vdots & \ddots & \vdots \\ (\varphi \circ \theta)^m_1 & \cdots & (\varphi \circ \theta)^m_l \end{pmatrix} = \begin{pmatrix} \varphi^1_1 & \cdots & \varphi^1_n \\ \vdots & \ddots & \vdots \\ \varphi^m_1 & \cdots & \varphi^m_n \end{pmatrix} \cdot \begin{pmatrix} \theta^1_1 & \cdots & \theta^1_l \\ \vdots & \ddots & \vdots \\ \theta^n_1 & \cdots & \theta^n_l \end{pmatrix} = \Phi \cdot \Theta.$$

5. Let $(V, \oplus, \odot)$ be a vector space and let $\varphi: V \rightarrow V^*$ and $\psi: V \rightarrow V$ be linear maps. Prove that the map

$$\begin{aligned} \pi: V \times V &\rightarrow \mathbb{R} \\ (v, w) &\mapsto \varphi(v)(\psi(w)) \end{aligned}$$

is a bilinear map on V . (Bilinear means independently linear in each slot.)

Solution: For a map to be linear, it must preserve the vector space addition and scalar multiplication. Let $v_1, v_2, w_1, w_2 \in V$ and $\lambda, \mu \in \mathbb{R}$. Then we must show that the four equalities hold:

$$\begin{aligned} \pi(v_1 \oplus v_2, w) &= \pi(v_1, w) + \pi(v_2, w), \\ \pi(v, w_1 \oplus w_2) &= \pi(v, w_1) + \pi(v, w_2), \\ \pi(\lambda \odot v, w) &= \lambda \cdot \pi(v, w), \\ \pi(v, \mu \odot w) &= \mu \cdot \pi(v, w). \end{aligned}$$

Let us denote the vector space structure of the dual space with $(V^*, \boxplus, \boxdot)$. Then

$$\begin{aligned} \pi(v_1 \oplus v_2, w) &= \varphi(v_1 \oplus v_2)(\psi(w)) && \text{definition of } \pi \\ &= (\varphi(v_1) \boxplus \varphi(v_2))(\psi(w)) && \text{linearity of } \varphi \\ &= \varphi(v_1)(\psi(w)) + \varphi(v_2)(\psi(w)) && \text{definition of } \boxplus \text{ in } V^* \\ &= \pi(v_1, w) + \pi(v_2, w) && \text{definition of } \pi. \end{aligned}$$

The other three equalities can be proven in a similar way.

6. Consider the vector space $(\mathbb{R}^2, +, \cdot)$ with the standard addition and scalar multiplication and two vectors $e_1 = (1, 2)$ and $e_2 = (3, 4)$.

(a) Is the map

$$\begin{aligned} f: \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x, y) &\mapsto 3x + 2y \end{aligned}$$

- a linear map? Of what space f is an element?
- (b) Check that the vectors e_1 and e_2 form a basis in $\mathbb{R}^2$.
- (c) Find the basis $(\varepsilon^1, \varepsilon^2)$ of $(\mathbb{R}^2)^*$ dual to the basis (e_1, e_2) .
- (d) Find the components of f in the dual basis, i.e. the numbers $f_1, f_2 \in \mathbb{R}$ such that $f = f_1 \cdot \varepsilon^1 + f_2 \cdot \varepsilon^2$.

Solution:

- (a) $f((a, b) + (c, d)) = f(a + c, b + d) = 3(a + c) + 2(b + d) = (3a + 2b) + (3c + 2d) = f(a, b) + f(c, d)$ and $f(\lambda(a, b)) = f(\lambda a, \lambda b) = 3\lambda a + 2\lambda b = \lambda(3a + 2b) = \lambda f(a, b)$.
So yes, f is linear. Since it maps to $\mathbb{R}$ it is a covector, i.e., $f \in (\mathbb{R}^2)^*$
- (b) e_1 and e_2 form a basis because they are linearly independent.
- (c) You have to find ϵ^i such that $\epsilon^i(e_j) = \delta_j^i$. This can be done by setting up the system on linear equations encoded by $(\alpha \ \beta) \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $(\bar{\alpha} \ \bar{\beta}) \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ where $\epsilon^1 = (\alpha, \beta)$ and $\epsilon^2 = (\bar{\alpha}, \bar{\beta})$. Solution is $\alpha = -2$, $\beta = \frac{3}{2}$, $\bar{\alpha} = 1$, $\bar{\beta} = -\frac{1}{2}$.
- (d) You let f act on the basis vectors e_i to obtain f_i

7. Consider the vector space $(V, +, \cdot)$, where V is the space of second order polynomials on the interval $[-1, 1]$, i.e. any element of V is of the form

$$f: [-1, 1] \rightarrow \mathbb{R}$$

$$x \mapsto ax^2 + bx + c$$

where $a, b, c \in \mathbb{R}$, and the operations are defined point-wise. Consider as basis vectors $e_1 = (x \mapsto x^2)$, $e_2 = (x \mapsto x)$ and $e_3 = (x \mapsto 1)$. The components of any vector $f(x) = ax^2 + bx + c$ with respect to the chosen basis are $f^1 = a$, $f^2 = b$ and $f^3 = c$.

- (a) Why is this vector space 3-dimensional?
- (b) Compute the dual basis $(\varepsilon^1, \varepsilon^2, \varepsilon^3)$ with respect to the chosen basis.
- (c) We define the *Dirac delta* $\bar{\delta}$ as the operator acting on an element of V and extracting its value at $x = 0$, i.e. $\bar{\delta}(f) := f(0)$. Show that $\bar{\delta}$ is a co-vector, i.e., $\bar{\delta} \in V^*$.
- (d) What is the component representation of $\bar{\delta}$ with respect to the previously computed dual basis?

Solution:

- (a) The basis has three elements

- (b) You have to find ϵ^i such that $\epsilon^i(e_j) = \delta_j^i$. $\epsilon^1(\cdot) = \frac{1}{2} \frac{\partial^2(\cdot)}{\partial x^2} \big|_{x=0}$, $\epsilon^2(\cdot) = \frac{\partial(\cdot)}{\partial x} \big|_{x=0}$, $\epsilon^3(\cdot) = (\cdot) \big|_{x=0}$
- (c) $\bar{\delta}$ is a linear map with domain V and target $\mathbb{R}$
- (d) $\bar{\delta} = \bar{\delta}_i \epsilon^i$, where $\bar{\delta}_1 = \bar{\delta}_2 = 0$ and $\bar{\delta}_3 = 1$